

of integers from
a to b inclusive is
 $b-a+1$

HW2 2.5, P. 161-2

16a. $\sum_{j=0}^8 (1+(-1)^j)$

When j is even, $(-1)^j = 1$, so $1+1=2$

When j is odd, $(-1)^j = -1$, so $1+(-1)=0$

Sum is 2 for even j and 0 for odd j .

Even j from 0 to 8 are 0, 2, 4, 6, 8 (5 terms).
 $\sum_{j=0}^8 (1+(-1)^j) = 5 \cdot 2 = 10$
 (5 indices)

16b. $\sum_{j=0}^8 (3^j - 2^j) = \sum_{j=0}^8 3^j - \sum_{j=0}^8 2^j$ Telescoping sum

$\sum_{j=0}^8 3^j = \frac{3^9 - 1}{3 - 1} = \frac{19683 - 1}{2} = \frac{19682}{2} = 9841$
 $\sum_{j=0}^8 2^j = \frac{2^9 - 1}{2 - 1} = 512 - 1 = 511$
 $\rightarrow 9841 - 511 = 9330$

* 26. $\sum_{j=0}^m \lfloor \sqrt[3]{j} \rfloor$, Let $K = \lfloor \sqrt[3]{j} \rfloor$. Then $K \leq \sqrt[3]{j} < K+1$, so
 $K^3 \leq j < (K+1)^3$
 Intervals (subtracted a from b) $[a, b]$

For each integer $K \geq 0$, the values of j for which $\sqrt[3]{j} = K$ are $j = K^3, K^3+1, \dots, (K+1)^3-1$

Numbers of j : $(K+1)^3 - K^3 = 3K^2 + 3K + 1$

$\sum_{j=0}^m \lfloor \sqrt[3]{j} \rfloor = \sum_{K=0}^{q-1} K \cdot (3K^2 + 3K + 1) + q \cdot (m - q^3 + 1)$, where $q = \lfloor \sqrt[3]{m} \rfloor$

$\sum_{K=0}^{q-1} K(3K^2 + 3K + 1) = \sum_{K=0}^{q-1} (3K^3 + 3K^2 + K)$ $\rightarrow$ equivalent expressions

$\sum_{K=0}^{q-1} K^3 = \left(\frac{(q-1)q}{2} \right)^2$ $\sum_{K=0}^{q-1} K^2 = \frac{(q-1)q(2q-1)}{6}$ $\sum_{K=0}^{q-1} K = \frac{(q-1)q}{2}$

$3 \left(\frac{(q-1)^2 q^2}{4} \right) + 3 \left(\frac{(q-1)q(2q-1)}{6} \right) + \frac{(q-1)q}{2}$
 $\rightarrow \frac{3(q-1)^2 q^2}{4} + \frac{(q-1)q(2q-1)}{2} + \frac{(q-1)q}{2}$

$\sum_{j=0}^m \lfloor \sqrt[3]{j} \rfloor = \frac{3(q-1)^2 q^2}{4} + \frac{(q-1)q(2q-1)}{2} + \frac{(q-1)q}{2} + q(m - q^3 + 1)$ where $K = \lfloor \sqrt[3]{m} \rfloor$